TITANIC SURVIVAL PREDICTION USING LOGISTIC REGRESSION

TEAM - 64

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Repo link- https://github.com/2026mcb1515-strgzr/AI_miniproject

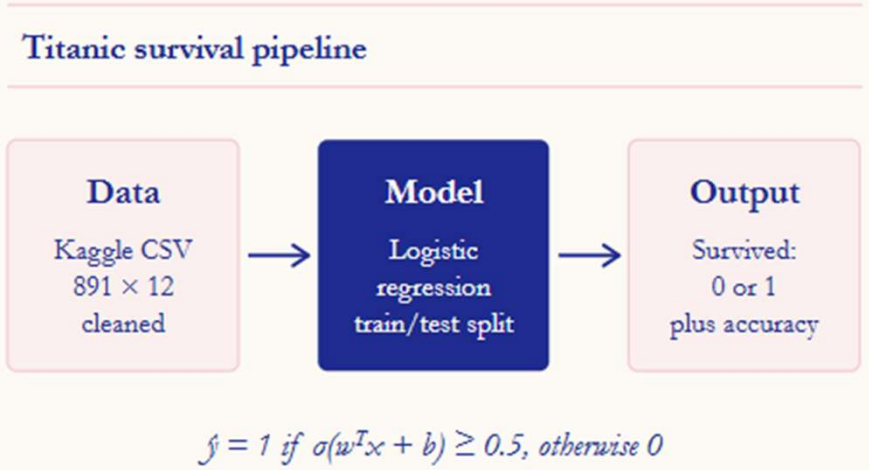
PROBLEM AND DATA

- The problem is to predict the survival of an individual on the titanic with binary classification using logistic regression(0=died, 1=survived).
- Logistic Regression is used because it is a simple interpretable baseline that shows how factors like age, sex and class affected survival on the Titanic.
- The data set is a .csv file obtained from Kaggle - https://www.kaggle.com/datasets/yasserh/titanic-dataset
- It consists of 12 columns: 11 input features plus the target(survival) and has 891 corresponding entries(one per passenger).

APPROACH

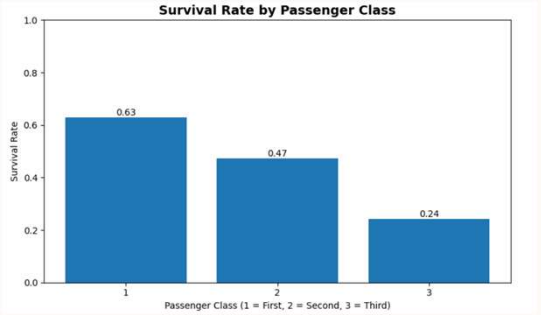
Libraries used:

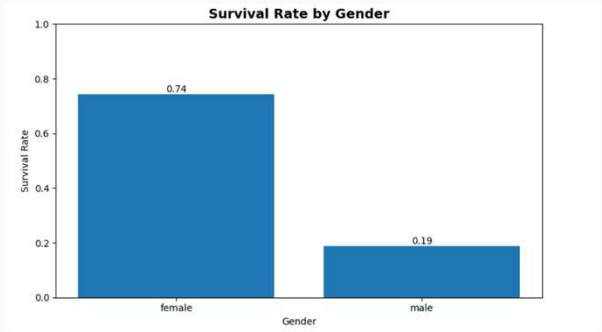
- Pandas: to load the csv file, and to inspect and clean data.
- Numpy: to deal with arrays and calculations.
- Scikit-learn: for the train-test split, logistic regression model and evaluation metrics.
- Matplotlib: for plots and charts
- Seaborn: for statistical plots
- Counter: for counting categories or class labels.



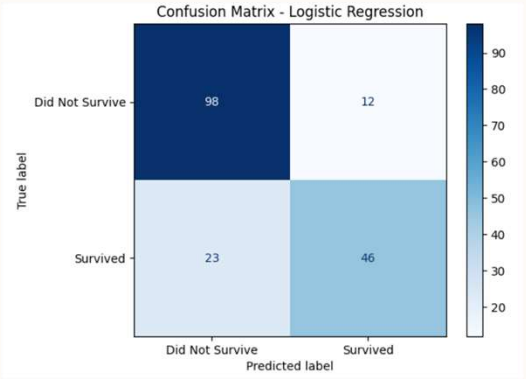
RESULTS

- The model's accuracy is 80.447%.
- Precision and Recall: 79.31% and 66.67%
- F1 Score: 72.44%





Confusion matrix



Coefficients and intercept of the model

	Feature	Coefficient
0	Pclass	-0.928698
1	Sex	1.269598
2	Age	-0.502878
3	SibSp	-0.262672
4	Parch	-0.067158
5	Fare	0.099599
6	Embarked_Q	0.080683
7	Embarked_S	-0.171784

Intercept: -0.657

Demo of the model

```
import pandas as pd

sample_passenger = X_test.iloc[[0]]

prediction = model.predict(sample_passenger)

original_passenger = df.loc[sample_passenger.index]

print("Passenger Details:")
print(original_passenger)

if prediction[0] == 1:
    print("Prediction: Survived")
else:
    print("Prediction: Did Not Survive")
```

```
Passenger Details:
     PassengerId  Survived  Pclass                 Name   Sex   Age  SibSp  \
565          566         0       3  Davies, Mr. Alfred J  male  24.0      2

     Parch     Ticket   Fare Cabin Embarked
565      0  A/4 48871  24.15   NaN        S
Prediction: Did Not Survive
```

LEARNINGS

- What worked:
    - Clean pipeline (data → model → output) made each step easy to check.
    - Fitting StandardScaler on the training data avoided test-set leakage.

- What didn't work:
    - Logistic regression assumes that log odds are a linear combination of the features, which may not be the case.
    - Smaller dataset due to which there are relatively noisier results.
- Who did what:
    - Aryan: Handled the github documentation and loaded the csv file along with required libraries.
    - Akshit: Made the requirements.txt file and preprocessed the data.
    - Goutham: Made the PPT and trained the model.
    - Maitrayee: Made the video and evaluated the model.
- AI disclosure: AI (Claude) was used to generate the pipeline diagram shown on slide 3. It was also used to create roadmap on how to approach the chosen problem.